

Ryo Wijaya

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SUMMARY

Software Engineer with 3+ years building distributed, cloud-native backend systems and event-driven ETL pipelines (Spring Boot, Kafka, Spark, Kubernetes, GCP) in fintech; skilled in microservices architecture, DevOps, and AI engineering.

EDUCATION

National University of Singapore (NUS) | Singapore

Aug. 2020 – June 2024

Bachelor of Computing, Information Systems (Hons. w/ Distinction) | FinTech Specialisation

EXPERIENCE

Software Engineer | Crédit Agricole Corporate & Investment Bank

July 2024 – Present

- As a founding engineer on a **greenfield project**, built modular **ETL Treatment Chains** using **Spark Streaming**, **Spring Boot**, and **Google Cloud Platform**; developed and maintained Java drivers across a suite of 22+ ingestion, enrichment, and distribution pipelines spanning Commercial Real Estate Loans, ESG, and Capital Markets data.
- Developed, tested, and shipped REST APIs and backend features across the core **Spring Boot** applications, delivering functionality spanning security integrations, configuration management, and **Kafka**-based messaging.
- Owner and main contributor (7 of 9 core modules) of an internal **Spring Boot** DevTools platform covering automated **JSON** ↔ **Avro** conversion, **Kafka** topic management, and **Apache Iceberg** DDL generation from Avro schemas.
- Developed a configuration validation and loading platform adopted across teams in Singapore and Paris to manage application inputs, reducing manual validation overhead and accelerating development and UAT cycles.
- Architected and implemented **OpenID Connect (OIDC)** and **OAuth2 SSO** authentication for the **Angular** application platform; also developed UI features.
- Engineered Java Avro deserializers and schema validation pipelines across both **Spring Kafka** and **Spark Kafka** consumers, integrating with Confluent Schema Registry to enforce strict schema contracts and eliminate non-compliant ingestion.
- Managed end-to-end Avro schema versioning cycles, propagating changes across pipeline components and coordinating upstream/downstream integration teams through SIT to ensure pipeline integrity for UAT and production releases.
- Designed and maintained **GitLab CI** pipelines covering automated build, test, **Docker** image builds, and staging/stable registry promotions; orchestrated application deployments using **Helm** and **Kubernetes**.
- Strengthened application security by remediating **Checkmarx** and **JSFrog Xray** vulnerabilities and enforcing **SonarQube** quality gates across all feature deployments and targeted large-scale remediation efforts.

Software Engineer Intern | SP Group

Jan. 2023 – June 2023

- Developed an **OCPP 1.6/2.0.1** EV charger simulator (**FastAPI**, **WebSockets**); load tested **2,000+** chargers.
- Built a **Next.js** platform for real-time charger configuration, message triggering, and log monitoring.

Software Engineer Intern | Reluvate Technologies

May 2022 – Aug. 2022

- Built a hiring platform (**Node**, **GCP**) automating **90%+** of recruitment: document exchange and assessments.
- Built a **Python serverless** backend for an accounting platform to handle payments and contracts.

PERSONAL SOFTWARE PROJECTS

Drift | *FastAPI, Next.js, Postgres, GCP* | AI-narrated portfolio digest for retail investors and traders

FinAI | *LangGraph, Qdrant, GCP* | Multi-tenant AI financial platform with KPI extraction, compliance gap analysis, and copilot **trade-signals** | *FastAPI, Telegram* | Telegram bot for daily stock technical analysis with AI-powered market summaries

PetHub | *Express.js, Next.js, PostgreSQL, Prisma, AWS* | Fully featured E-commerce pet-service marketplace

ryo & blog | *React, JavaScript, Vite* | Portfolio website and tech blog

SKILLS

Languages: Java, Python, JavaScript/TypeScript, HTML, SQL

Technologies: Spring Boot, Apache Kafka (Confluent), Apache Spark, Apache Iceberg, Avro, ETL, RESTful APIs, FastAPI, Next.js, React.js, Node.js, Angular, PostgreSQL, Jakarta EE, JUnit, LangGraph, LangChain, Qdrant, Ollama

Infrastructure & Tools: DevOps, Kubernetes, Helm, Docker, Containerization, GitLab CI, CI/CD, Git, Maven, Linux, Google Cloud Platform (GCP), GKE, GCS, Cloud Run, Azure, SonarQube, Checkmarx, JSFrog Xray

Others: Microservices Architecture, Event-Driven Architecture, Distributed Systems, OIDC/OAuth2, RAG, Agile, Scrum

Communication: English (Native), German (Conversational), Mandarin (Conversational)

CERTIFICATIONS

Certified Kubernetes Application Developer (CKAD) - 98/100 | *Cloud Native Computing Foundation*

2025

Confluent Apache Kafka Fundamentals Accreditation | *Confluent*

2026

Financial Technology Specialisation (Distinction) | *National University of Singapore*

2024